



Spotilytics

Data Validation & Quality Assurance Certificate — AI-620 Fundamentals of Data Engineering

Generated: April 30, 2026 | Dataset: Kaggle spotify-1million-tracks | Group 6

✓ GREAT EXPECTATIONS CERTIFIED

RAW TRACKS

1,159,764

Before cleaning

CLEAN TRACKS

1,155,971

After cleaning

DATA RETAINED

99.67%

Only 0.33% removed

YEARS COVERED

24

2000 – 2023

END-TO-END PIPELINE



Ingest
Kaggle API



Validate
Great Expectations



Clean
PySpark



Store
PostgreSQL



ML + API
MLlib + FastAPI

DATA CLEANING SUMMARY

BEFORE CLEANING

1,159,764

Raw CSV rows

AFTER CLEANING

1,155,971

Clean Parquet rows

ROWS REMOVED

3,793

0.33% of total

- Duplicate track_ids removed ·2,341 rows
- Null artist/track names dropped ·1,205 rows
- Outliers filtered (tempo=0, loudness≥0, duration<30s) ·247 rows
- Bad values fixed (key=-1 → null) 0 rows removed

Data retention rate 99.67%

STAGE 1 — RAW DATA VALIDATION (PRE-CLEANING)

Raw CSV — 18 checks · 1,159,764 rows

Failures expected — fixed by PySpark cleaning pipeline · Timestamp: 2026-04-30 04:56:58

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required_columns_present	PASS	row_count_above_1m	PASS
track_id_not_null	PASS	track_id_unique	PASS
artist_name_not_null	FAIL	track_name_not_null	FAIL
popularity_range_0_100	PASS	year_range_2000_2023	PASS
danceability_range_0_1	PASS	energy_range_0_1	PASS
valence_range_0_1	PASS	acousticness_range_0_1	PASS
speechiness_range_0_1	PASS	instrumentalness_range_0_1	PASS
liveness_range_0_1	PASS	tempo_positive	FAIL
loudness_negative	FAIL	duration_ms_above_30s	FAIL

STAGE 2 — CLEANED DATA VALIDATION (POST PYSPARK)

Cleaned Parquet — 16 checks · 1,155,971 rows

All checks must pass — pipeline halts on any failure · Timestamp: 2026-04-30 05:00:48

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row_count_above_1m	PASS	no_duplicate_track_ids	PASS
track_id_not_null	PASS	artist_name_not_null	PASS
track_name_not_null	PASS	genre_not_null	PASS
popularity_not_null	PASS	popularity_range_0_100	PASS
year_range_2000_2023	PASS	danceability_range_0_1	PASS
energy_range_0_1	PASS	tempo_positive	PASS
loudness_negative	PASS	duration_ms_above_30s	PASS
genre_group_column_present	PASS	popularity_tier_column_present	PASS

STAGE 3 — ML MODEL EVALUATION · BOPORFLOP · TIMESTAMP: 2026-04-30 05:03:46

F1 SCORE	ACCURACY	PRECISION	RECALL
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0.7516

79.98%

0.7664

0.7998

FEATURE IMPORTANCE

year	44.8%
genre_idx	17.9%
duration_ms	9.6%
instrumentalness	6.3%
danceability	5.0%
loudness	4.4%
acousticness	4.2%
valence	2.1%
energy	2.0%
speechiness	1.7%
liveness	1.5%
tempo	0.4%

CONFUSION MATRIX (LOW=0, MID=1, HIGH=2)

ACTUAL / PRED	LOW (0)	MID (1)
Low (0)	174,814	4,040
Mid (1)	39,349	9,864
High (2)	1,601	1,237

LABEL DISTRIBUTION (REAL DATA)

Low	77.5%
Mid	21.3%
High	1.2%

Train: 925,066 rows | Test: 230,905 rows | Split: 80/20

MODEL CONFIGURATION

ALGORITHM

Random Forest

TREES

100

MAX DEPTH

10

FEATURES

12

Spotilytics – AI-620 Data Engineering Pipeline

Kaggle → PySpark → Great Expectations → PostgreSQL → MLlib → FastAPI → React → Deployed on Render + Vercel

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✓ DATA QUALITY CERTIFIED – 16/16 CHECKS PASSED